

ASHISH KUMAR YADAV

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Professional Summary

Enthusiastic and motivated undergraduate Civil Engineering student with a strong interest in computational mechanics, finite element analysis, and physics-informed machine learning. Eager to contribute to research and engineering projects while developing advanced technical skills.

Education

Maulana Azad National Institute of Technology (MANIT), Bhopal
B.Tech in Civil Engineering (Expected Aug 2028)
SGPA: Sem 1: 8.50 | Sem 2: 8.85 | Sem 3: 9.50 | Current CGPA: 8.94

Kendriya Vidyalaya, Mau
Class XII (2023): 88.8%
Class X (2021): 97.4%

Skills

• MATLAB • Python • Finite Element Method (FEM) • Physics-Informed Neural Networks (PINNs) • LibreCAD • Problem Solving • Teamwork

Projects

- **PINNs for 1D Heat Equation:** Developed a Physics-Informed Neural Network in MATLAB to solve the transient one-dimensional heat equation by enforcing governing PDE, boundary conditions, and initial conditions.
- **PINNs for Euler–Bernoulli Beam:** Implemented a PINN-based framework in MATLAB to predict beam deflection by satisfying the Euler–Bernoulli governing equation and boundary conditions.
- **Finite Element Analysis of Beams:** Developed MATLAB codes for Euler–Bernoulli and Timoshenko beam formulations, including stiffness matrix implementation, numerical integration, and structural response analysis.

Achievements & Certifications

- Participated in an AutoCAD Workshop.
- Presented a paper at the 1st National Conference GEODCON 2026: “Selection of map projection for large scale mapping in India: A comparative analysis of LCC and other projections with UTM.”